

KAMEN-NA-OBI, Russian Federation

WMO: 298220

Lat: 53.8189N

Lon: 81.2768E

Elev: 424

StdP: 14.47

Time Zone: 7.00 (E07)

Period: 94-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.5	-29.1	-40.2	0.6	-35.2	-33.8	0.8	-28.7	23.7	20.7	21.8	18.5	2.2	250	0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.3	87.3	65.7	84.0	64.9	81.0	63.9	69.3	80.8	67.8	78.9	66.2	77.2	8.5	70

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.7	96.4	74.6	63.8	89.9	72.7	62.2	84.9	71.3	33.8	80.8	32.5	78.9	31.2	77.3	77.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
21.2	18.3	15.8	DB	-40.9	92.9	7.8	3.9	-46.5	95.7	-51.1	98.0	-55.4	100.2	-61.1	103.0
			WB	-40.2	72.6	7.3	2.4	-45.4	74.3	-49.7	75.7	-53.8	77.1	-59.1	78.8

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	35.7	-1.9	3.7	18.9	39.8	54.0	64.8	67.5	63.2	51.4	38.1	20.7	6.4
	DBStd	26.65	16.32	13.71	12.21	10.30	9.61	7.48	5.89	6.61	8.38	9.26	13.94	15.13
	HDD50	6976	1609	1298	964	335	73	4	0	1	81	381	878	1352
	HDD65	10976	2074	1718	1429	757	356	93	42	114	414	834	1328	1817
	CDD50	1759	0	0	0	28	197	448	542	409	123	12	0	0
	CDD65	286	0	0	0	0	16	87	119	57	7	0	0	0
	CDH74	3084	0	0	0	16	276	915	1070	670	137	0	0	0
	CDH80	865	0	0	0	2	62	258	316	187	40	0	0	0
(14) Wind	WSAvg	6.3	6.2	6.3	6.8	7.4	7.5	5.8	4.7	5.0	5.5	6.5	7.2	7.3
(15) Precipitation	PrecAvg	14.0	0.6	0.6	0.6	0.8	1.2	1.7	2.6	1.5	1.4	1.1	1.1	0.9
	PrecMax	18.4	1.3	1.5	1.4	1.9	3.0	3.4	6.2	2.8	3.1	2.8	2.0	1.9
	PrecMin	8.9	0.1	0.0	0.0	0.0	0.1	0.4	0.7	0.4	0.1	0.4	0.5	0.4
	PrecStd	2.8	0.3	0.3	0.4	0.5	0.7	0.9	1.3	0.7	0.8	0.7	0.4	0.4
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	34.4	46.2	76.1	87.2	89.9	91.5	90.5	85.7	68.5	48.8	35.3
		MCWB	32.6	32.3	39.3	55.9	63.1	65.5	68.0	67.9	62.9	53.6	43.4	33.2
	2%	DB	29.6	30.8	40.6	68.4	81.2	86.2	87.0	84.9	77.4	62.2	44.2	32.7
		MCWB	27.9	29.3	36.1	52.8	59.7	64.4	67.1	66.4	59.5	50.2	40.0	31.1
	5%	DB	25.6	27.6	37.3	62.2	77.1	82.7	83.4	81.0	72.0	57.0	40.0	29.6
		MCWB	24.2	26.1	34.1	49.2	58.0	64.1	66.4	64.2	57.2	47.9	37.3	28.0
	10%	DB	21.9	24.0	34.8	56.4	72.1	79.5	80.1	77.2	66.8	51.5	36.5	26.3
		MCWB	20.5	22.7	32.3	46.5	56.2	63.2	65.5	63.1	55.0	45.1	34.3	24.9
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.7	33.0	39.9	57.4	65.9	70.0	72.1	70.8	64.7	54.6	44.6	33.3
		MCDB	33.6	34.1	44.9	72.7	82.1	82.3	83.7	82.0	80.6	63.9	48.5	34.6
	2%	WB	28.1	29.4	36.6	53.8	62.4	67.7	70.1	68.2	60.8	51.3	40.2	31.3
		MCDB	29.4	30.7	39.9	66.6	76.5	79.6	81.0	80.5	74.3	60.0	43.2	32.7
	5%	WB	24.3	26.3	34.6	50.4	59.5	66.0	68.5	66.2	58.3	48.5	37.5	28.2
		MCDB	25.5	27.6	37.2	61.0	72.9	78.3	78.9	77.6	69.5	55.7	40.3	29.5
	10%	WB	20.6	22.7	32.4	46.8	57.3	64.6	67.1	64.2	55.8	45.6	34.3	24.8
		MCDB	21.9	24.1	34.6	56.2	70.4	76.1	77.1	74.9	65.4	51.7	36.3	26.3
(35) Mean Daily Temperature Range	5% DB	MDBR	16.7	18.4	18.2	19.1	22.3	20.6	19.3	21.9	21.3	16.1	12.1	14.7
		MCDBR	15.5	15.2	16.5	28.9	29.7	26.5	24.8	28.2	30.9	26.4	12.7	14.9
		MCWBR	15.4	14.6	13.0	15.4	13.3	10.7	9.3	12.0	14.6	15.5	9.8	14.5
		MCDBR	15.7	15.3	14.6	27.6	25.6	21.6	20.3	23.8	26.3	24.1	12.3	14.6
	5% WB	MCWBR	15.5	15.0	11.9	15.6	13.1	10.0	9.3	10.6	13.6	14.9	9.9	14.4
(40) Clear-Sky Solar Irradiance	taub		0.298	0.331	0.384	0.411	0.413	0.411	0.437	0.416	0.374	0.356	0.335	0.302
	taud		2.013	1.957	1.902	2.114	2.191	2.283	2.210	2.243	2.306	2.234	2.064	1.965
	Ebn at Noon		195	228	242	257	265	266	256	254	250	223	178	161
	Edh at Noon		24	36	47	44	43	40	42	38	32	27	23	21
(44) All-Sky Solar Radiation	RadAvg		339	677	1143	1500	1797	1957	1762	1516	1061	554	271	227
	RadStd		29	42	92	96	170	156	115	108	109	74	44	33

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (15 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page